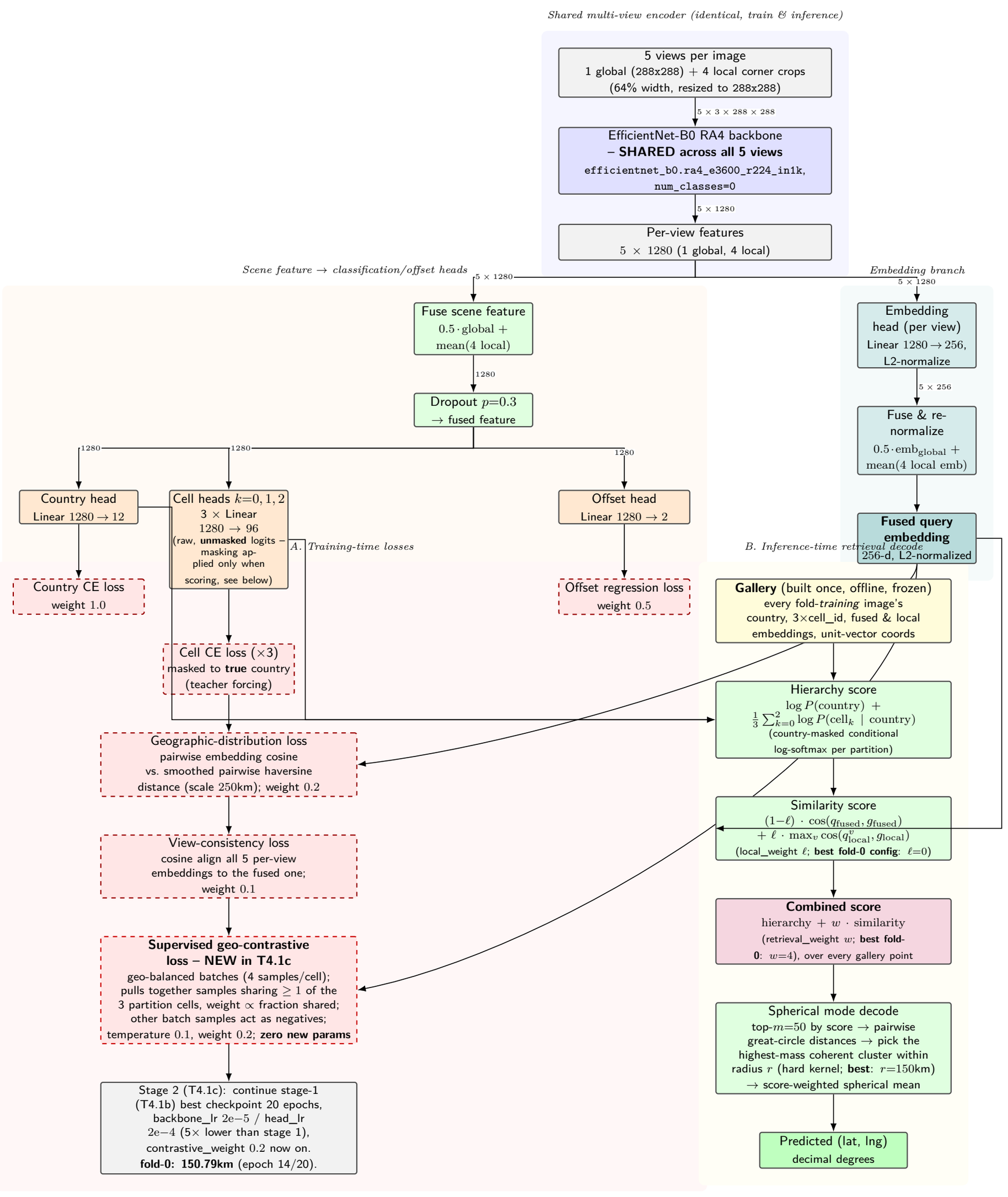
Shared multi-view encoder (identical, train & inference)
5 views per image
1 global (288x288) + 4 local corner crops
(64% width, resized to 288x288)
5 × 3 × 288 × 288
EfficientNet-B0 RA4 backbone
– SHARED across all 5 views
efficientnet_b0.ra4_e3600_r224_in1k,
num_classes=0
5 × 1280
Per-view features
5 × 1280 (1 global, 4 local)
Scene feature → classification/offset heads
5 × 1280
Fuse scene feature
0.5 · global +
mean(4 local)
1280
Dropout p=0.3
→ fused feature
1280
1280
1280
Country head
Linear 1280 → 12
Cell heads k=0, 1, 2
3 × Linear
1280 → 96
(raw, unmasked logits – masking applied only when scoring, see below)
Offset head
Linear 1280 → 2
A. Training-time losses
Country CE loss
weight 1.0
Cell CE loss (×3)
masked to true country
(teacher forcing)
Offset regression loss
weight 0.5
Geographic-distribution loss
pairwise embedding cosine
vs. smoothed pairwise haversine
distance (scale 250km); weight 0.2
View-consistency loss
cosine align all 5 per-view
embeddings to the fused one;
weight 0.1
Supervised geo-contrastive
loss – NEW in T4.1c
geo-balanced batches (4 samples/cell);
pulls together samples sharing ≥ 1 of the
3 partition cells, weight ∝ fraction shared;
other batch samples act as negatives;
temperature 0.1, weight 0.2; zero new params
Stage 2 (T4.1c): continue stage-1
(T4.1b) best checkpoint 20 epochs,
backbone_lr 2e−5 / head_lr
2e−4 (5× lower than stage 1),
contrastive_weight 0.2 now on.
fold-0: 150.79km (epoch 14/20).
Embedding branch
5 × 1280
Embedding
head (per view)
Linear 1280 → 256,
L2-normalize
5 × 256
Fuse & re-
normalize
0.5 · emb_global +
mean(4 local emb)
Fused query
embedding
256-d, L2-normalized
B. Inference-time retrieval decode
Gallery (built once, offline, frozen)
every fold-training image's
country, 3×cell_id, fused & local
embeddings, unit-vector coords
Hierarchy score
log P(country) +
1/3 Σ_{k=0}^{2} log P(cell_k | country)
(country-masked conditional
log-softmax per partition)
Similarity score
(1−ℓ) · cos(q_fused, g_fused)
+ ℓ · max_v cos(q^v_local, g_local)
(local_weight ℓ; best fold-0 config: ℓ=0)
Combined score
hierarchy + w · similarity
(retrieval_weight w; best fold-
0: w=4), over every gallery point
Spherical mode decode
top-m=50 by score → pairwise
great-circle distances → pick the
highest-mass coherent cluster within
radius r (hard kernel; best: r=150km)
→ score-weighted spherical mean
Predicted (lat, lng)
decimal degrees